Estimating Pi

Chris Webb

July 28, 2026

$$\pi \approx 3.141592653589793238$$

1 Fractions

Default vertical spacing :(

$$\pi \approx \frac{22}{7}$$
$$\pi \approx \frac{333}{106}$$
$$\pi \approx \frac{355}{113}$$
$$\pi \approx \frac{52163}{16604}$$
$$\pi \approx \frac{103993}{33102}$$
$$\pi \approx \frac{245850922}{78256779}$$

vertical spacing 1.0ex :)

$$\pi \approx \frac{22}{7}$$

$$\pi \approx \frac{333}{106}$$

$$\pi \approx \frac{355}{113}$$

$$\pi \approx \frac{52163}{16604}$$

$$\pi \approx \frac{103993}{33102}$$

$$\pi \approx \frac{245850922}{78256779}$$

2 Francois Viete

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdots$$

$$\pi = (1 \div \frac{2}{\pi}) \times 2$$

$$\pi = \left(1 \div \frac{2}{\pi}\right) \times 2$$

3 John Wallis

$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \frac{8}{7} \cdot \frac{8}{9} \cdots$$

4 John Machin

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239}$$

5 Gregory-Leibniz

$$\pi = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + \frac{4}{13} - \cdots$$

6 Nilakantha

$$\pi = 3 + \frac{4}{2 \times 3 \times 4} - \frac{4}{4 \times 5 \times 6} + \frac{4}{6 \times 7 \times 8} - \frac{4}{8 \times 9 \times 10} + \cdots$$